

# USE TECHNOLOGY TO GROW YOUR OWN SUSTAINABLE GARDEN IN A SMALL SPACE

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(Credit: hireplants.com)



Vertical gardening in office spaces  
(Credit: [www.kickstarter.com](http://www.kickstarter.com))

**D**o you wish to have a garden in your home? Is it space that is stopping you? Here is a solution for you—Internet of Things (IoT)-based vertical gardening. This technology helps you utilise your walls to grow a vertical garden. It is basically a hydroponic garden that helps you grow plants without soil.

Plants require sunlight, water and nutrients from soil. In this case, soil is replaced with a nutrient-rich water solution to fulfill that requirement. The innovation lets you grow a garden anywhere, including supermarkets, apartment complexes, office spaces, restaurants and hospitals.

This automated technology is powered by the IoT. The IoT is a network of physical objects—vehicles, devices, buildings and other items—embedded with electronics, sensors and software. Network connectivity enables these objects to

collect and exchange data. It monitors and keeps track of the environmental conditions in your garden and gives recommendations regarding plants that would do best in your garden based on soil composition.

There is a wide scope for growing plants in this manner, and you can select which plants you would like to grow. IoT-based vertical gardening gives high density yields and short growth cycles with reduced resource consumption of water, fertiliser and space. It saves labour costs, and offers higher consistency of